Scenario 3: Retrieving a List of Products

Step 1: Set Up the Project

- Initialize a new project
- Install necessary packages for building the API
- Create an array containing product details. Each product object should include properties like id, name, category, price, etc.

Step 2: Implement Routes and Controller Functions

Create routes and corresponding controller functions for the following endpoints:

- **GET /products**– Get list of all products.
- **GET /products/:id** – Retrieves detailed information for a specific product by its ID.
- **POST /products** – Creates a new product entry.
- **PUT /products/:id** – Updates details of an existing product.
- **PATCH /products/:id** – Update specific details of an existing product.
- **DELETE /products/:id** – Deletes a specific product by its ID.

Step 3: Accept Product ID, Category as a URL Query Parameter

- Implement logic to accept a product ID as a URL query parameter and send the details of the product corresponding to that ID.
- Implement logic to accept a product category as a URL query parameter and send the details of products matching the specified category.

Step 4: Accept URL Query Parameters for Product Properties and Sorting Order

- Implement logic to accept URL query parameters for product properties (e.g., category, price) and sorting order (e.g., asc or desc).
- Sort the product details based on the given property and sorting order.

Step 5: Validate Query Parameters

- Meticulously validate each query parameter to ensure accuracy and completeness before processing the request.
- Check for valid product IDs, existing categories, and proper sorting parameters.
- Handle errors gracefully if invalid or incomplete parameters are provided.